

LIU, Dingdong

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RESEARCH SUMMARY

Human-computer interaction (HCI) researcher building conversational AI systems that scaffold patient-clinician communication in healthcare, utilizing text- and voice-based agents to embodied humanoid robots, and spanning asynchronous telehealth, hospital admission interviews, neurocognitive disorder screening, and robot-assisted geriatric care. Author of **11 peer-reviewed publications** at ACM CHI, UIST, DIS, CSCW (PACM HCI), IEEE TVCG (VIS), and IEEE RO-MAN, including **5 as first or co-first author papers**. Among them one papers on humanoid-robot dialogue systems won the UBTECH Best Industry Application Award (Bronze) at IEEE RO-MAN 2024. Co-inventor on a US provisional patent; contributor to projects under the RGC Theme-based Research Scheme (TRS) and Areas of Excellence (AoE) scheme; recognized three times for outstanding reviewing at ACM CHI and DIS.

EDUCATION

The Hong Kong University of Science and Technology (HKUST), Hong Kong, China Doctor of Philosophy in Computer Science and Engineering Advisors: Prof. Xiaojuan Ma and Prof. Fugee Tsung	09/2022 – 09/2026 (expected) CGA: 3.85/4.30
• Expected degree conferral: Oct 15 th , 2026	
The Hong Kong University of Science and Technology (HKUST), Hong Kong, China Bachelor of Science in Computer Science Bachelor of Science in Data Science and Technology (Second Major) Graduated with First Class Honours	09/2018 – 06/2022 CGA: 3.84/4.30 MCGA: 3.89/4.30

PUBLICATIONS

† Equal contribution (co-first authorship)

UIST 2026 CoNarrate: Computationally Modeling Information Prematurity and Scaffolding the Co-Construction of Patient Narratives in Asynchronous Telehealth	05/2025 – 04/2026
• Dingdong Liu , Yun Wang, Hei Ching Iu, Li Zhang, Chuhan Shi, Yan Lu, Xiaojuan Ma. • To appear (Oct 2026) in <i>Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2026)</i> as first author . DOI: assigned, pending activation. • Work initiated during an internship at Microsoft Research Asia.	
TOCHI Dissecting Interaction Gulfs: Towards Automating Neurocognitive Disorder Screening Dialog Tasks	11/2024 – 04/2026
• Dingdong Liu , Lueyang Zhang, Junze Li, Hei Ching Iu, Bolin Zhao, Ka Ho Wong, Helen Meng, Xiaojuan Ma, Jiaxiong Hu. • Submitted to <i>ACM Transactions on Computer-Human Interaction</i> as first author . Under Review.	
CHI 2025 Scaffolded Turns and Logical Conversations: Designing Humanized LLM-Powered Conversational Agents for Hospital Admission Interviews	05/2024 – 09/2024
• Dingdong Liu , Yujing Zhang, Bolin Zhao, Shuai Ma, Chuhan Shi, Xiaojuan Ma. • Published in <i>Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI 2025)</i> , as first author . DOI: https://dl.acm.org/doi/10.1145/3706598.3714196	
RO-MAN 2025 Dynamic Prompting Improves Turn-taking in Embodied Spoken Dialogue Systems	11/2024 – 04/2025
• Yifan Shen†, Dingdong Liu †, Xiaoyu Mo, Fugee Tsung, Xiaojuan Ma, Bertram E. Shi. • Published in <i>IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2025)</i> as co-first author . DOI: https://doi.org/10.1109/RO-MAN63969.2025.11217726	
RO-MAN 2024 A Humanoid Robot Dialogue System Architecture Targeting Patient Interview Tasks	06/2023 – 08/2024
• Yifan Shen†, Dingdong Liu †, Yejin Bang†, Ho Shu Chan, Rita Frieske, Hoo Choun Chung, Jay Nieves, Tianjia Zhang, Kien T. Pham, Wai Yi Rosita Cheng, Yini Fang, Qifeng Chen, Pascale Fung, Xiaojuan Ma, Bertram E. Shi. • Published in <i>IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2024)</i> as co-first author . DOI: https://doi.org/10.1109/RO-MAN60168.2024.10731285 • UBTECH Best Industry Application Award – Bronze	
DIS 2024 PWIP Exploring Scaffolding Techniques for Agent-Administered Brief Cognitive Screening in Hospital Settings	06/2022 – 07/2023
• Dingdong Liu , Sensen Gao, Zixin Chen, Yifan Shen, Chuhan Shi, Bertram E. Shi, Xiaojuan Ma. • Published in <i>Companion Publication of the 2024 ACM Designing Interactive Systems Conference (DIS 2024, Provocations and Works-in-Progress)</i> as first author . DOI: https://doi.org/10.1145/3656156.3663697	

ADDITIONAL CO-AUTHORED PUBLICATIONS

- Jiaxiong Hu, Xiwen Yao, Zeyu Huang, Danxuan Liang, Dongjie Yang, **Dingdong Liu**, Junze Li, Yuanhao Zhang, Xiaojuan Ma. From Human Pragmatic Language Skills to Conversational Agent Design: A Systematic Review of Transfer Strategies. *Proc. ACM CHI 2026*. 2022 – 2026
- Junze Li, Yue Zhang, Chengbo Zheng, **Dingdong Liu**, Zeyu Huang, Xiaojuan Ma. InsightBridge: Enhancing Empathizing with Users through Real-Time Information Synthesis and Visual Communication. *Proc. ACM CHI 2025*.
- Zixin Chen, Jiachen Wang, Meng Xia, Kento Shigyo, **Dingdong Liu**, Rong Zhang, Huamin Qu. StuGPTViz: A Visual Analytics Approach to Understand Student-ChatGPT Interactions. *IEEE TVCG 31(1)*, 2025 (presented at IEEE VIS 2024).
- Chengzhong Liu, Shixu Zhou, **Dingdong Liu**, Junze Li, Zeyu Huang, Xiaojuan Ma. CoArgue: Fostering Lurkers' Contribution to Collective Arguments in Community-based QA Platforms. *Proc. ACM CHI 2023*.
- Chengzhong Liu, Shixu Zhou, Yuanhao Zhang, **Dingdong Liu**, Zhenhui Peng, Xiaojuan Ma. Exploring the Effects of Self-Mockery to Improve Task-Oriented Chatbot's Social Intelligence. *Proc. ACM DIS 2022*.
- Chengzhong Liu, Zeyu Huang, **Dingdong Liu**, Shixu Zhou, Zhenhui Peng, Xiaojuan Ma. PlanHelper: Supporting Activity Plan Construction with Answer Posts in Community-based QA Platforms. *Proc. ACM Hum.-Comput. Interact. 6(CSCW2)*, 2022.

PATENT

- USPTO | An Automated Scaffolding System for Neurocognitive Disorder Screening Dialog Tasks** 07/2025
- US Provisional Patent Application No. 63/854,195, filed July 30, 2025, USPTO (Patent Center #71603362)
 - Inventors: Helen Mei Ling Meng, Xiaojuan Ma, Jiaxiong Hu, **Dingdong Liu**, Ka Ho Wong.
 - Co-assigned to HKUST and CUHK.

RESEARCH EXPERIENCE

- Visiting Student Researcher** | Pervasive HCI Group, Tsinghua University 01/2026 – 04/2026
- Main Role: Led an independent project on post-dentist-care service design; Analysed user patterns and designed working strategies for clinician-patient-agent collaboration. Resulted in a first-author paper to be submitted to CHI 2027.
 - Host: Prof. [Yuntao Wang](#)
- Research Intern** | Microsoft Research Asia, Intelligent Multimedia Group 05/2025 – 09/2025
- Main Role: Led an independent project to improve telehealth communication: analysed 5,000+ real-world encounters, developed a taxonomy of information-delivery failures, and built a system that accelerated communication with improved information accuracy. Resulted in a first-author UIST 2026 paper.
 - Mentor: [Yun Wang](#)

GRANT-FUNDED RESEARCH INVOLVEMENT

- Research and Development of Artificial Intelligence in Extraction and Identification of Spoken Language Biomarkers for Screening and Monitoring of Neurocognitive Disorders** | Funded by RGC, T45-407/19N-2 06/2024 – 08/2025
- Main Role: Developed an intelligent screening system for cognitive impairment focusing on fully automated screening assessment and spoken-language biomarker extraction.
 - Work under this project led to the US provisional patent and the first-author manuscript under review at TOCHI.
- Frontier and Human-centric AI and Robotics Technology for Geriatric Care** | Funded by RGC, AoE/E-601/24-N 09/2022 – 08/2025
- Main Role: Designed and developed human-robot conversational systems for elderly care.
 - Work under this project led to two co-first-author IEEE RO-MAN papers (2024, award-winning; 2025) and a first-author ACM CHI 2025 paper.

TEACHING AND MENTORING

- Research Mentoring | HKUST Undergraduate Research Opportunities Program (UROP)** 09/2024 – 08/2025
- Mentored three undergraduate researchers across three UROP semesters (Fall 2024, Spring 2025, Summer 2025) on independent research in conversational agents for healthcare.
 - Mentees' work contributed to the TOCHI manuscript under review (Fall – Spring cohort) and to the user studies of the first-author UIST 2026 paper (Summer cohort), and produced a university-exhibited poster: “Evaluating Effects of Chain-of-Thought Strategies on LLM Performance as a Conversational Agent for Neurocognitive Disorder Screening.”
- Research Mentoring | Visiting Exchange Student** 09/2024 – 12/2024
- Mentored a visiting exchange student from the Technical University of Munich (TUM) in Germany on integrating clinical inference capabilities into LLM-based healthcare agents.
 - This early-stage exploration informed the direction of the first-author UIST 2026 paper.

Teaching Assistant | HKUST (four appointments)

2023 – 2025

- COMP1023 Introduction to Python Programming (Fall 2025) – **one of the lead TAs for the course's first offering** (600+ students): co-created the tutorial series used across all 10 weekly sessions and taught a weekly section of 60+ students for the full semester.
- Also: COMP2211 Exploring Artificial Intelligence (Spring 2025); COMP1022P Introduction to Computing with Java (Fall 2023); COMP4461 Human-Computer Interaction & CSIT6000N Advanced Topics in HCI (Spring 2023)

INVITED TALKS AND PRESENTATIONS

- Conversational Agents and Healthcare (invited talk), HKUST Engineering Summer Camp for Elite Students. Represented the Department of Computer Science and Engineering. 07/2026
- Conversational Agents and Human-Agent Interaction (invited talk), Central Academy of Fine Arts, Beijing. 06/2026
- Conference presentations: ACM CHI 2025 (paper, Yokohama); HHME 2026 (poster, Changsha); HHME 2025 (paper, Dalian); ACM DIS 2024 (poster, Copenhagen). 2024 – 2026

ACADEMIC SERVICE**Reviewer** 2024 – 2026

- ACM Conference on Human Factors in Computing Systems (CHI) | 2024, 2025, 2026
- ACM Conference on Human Factors in Computing Systems (CHI) Late-Breaking Work | 2024, 2025
- ACM Conference on Designing Interactive Systems (DIS) | 2025, 2026
- ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW) | 2024

Service Awards & Recognitions

2025 – 2026

- Special Recognition for Outstanding Review: CHI 2026, DIS 2025, CHI 2025 Late Breaking Work

LEADERSHIP**Person in Charge, Peer Mentor Program** | Mainland Students & Scholars Society (UG Section), HKUST 08/2019 – 09/2020

- Led a 10-member organizing committee and 100+ senior-student mentors to run a year-long peer-mentor program hosting 300+ first-year students from mainland China, Hong Kong, Macau, and overseas: an opening ceremony, an intensive two-week summer orientation of four events (ice-breaking game night, campus exploration, Hong Kong city tour, drama competition), and a closing ceremony. Owned overall program quality, timeline coordination, and delegation to per-event activity leads.

General Secretary | Mainland Students & Scholars Society (UG Section), HKUST 02/2019 – 02/2020

HONORS AND AWARDS**UBTECH Best Industry Application Award (Bronze)** | IEEE RO-MAN 2024 2024**RedBird Academic Excellence Award** | The Hong Kong University of Science and Technology 2024, 2025, 2026

- HKD 20,000/year

Kerry Holdings Limited Scholarship | The Hong Kong University of Science and Technology 2018 – 2022

- HKD 70,000/year

Dean's List (top 10% of students) | The Hong Kong University of Science and Technology 2018 – 2022
Five semesters